Experimental Evaluation of Multi-Rotor Aerodynamic Interactions

Abraham Atte*
Graduate Research Assistant
Georgia Institute of Technology
Atlanta, GA 30332

Daley Wylie*
Graduate Research Assistant
Georgia Institute of Technology
Atlanta, GA 30332

Juergen Rauleder
Assistant Professor
Georgia Institute of Technology
Atlanta, GA 30332

ABSTRACT

An investigation was performed into the aerodynamic interactions between the rotors of multi-rotor vehicles in different configurations. The effects of these interactions on the thrust and torque of all individual rotors were quantified in wind tunnel tests. Flow visualization and a limited number of particle image velocimetry measurements were conducted to get further insight into the aerodynamic interactions. The effects of the changes in hub spacings, rotor rotational speeds, and wind speeds were investigated for isolated, tandem, quad-rotor plus and X configurations. The maximum and minimum tip chord Reynolds numbers were 118,000 and 73,000, respectively. Results showed that the aft rotors experienced detrimental aerodynamic interactions in all configurations. In all examined multi-rotor configurations, an increase in the hub spacing caused a decrease in the thrust deficit between the aft rotor and the isolated rotor. However, the differences in the configurations also affected the measured loads. In the tandem configuration, the aft rotor experienced up to 24% reduction in the thrust coefficient at a hub spacing of $2.1R$ when compared to the isolated rotor at the same rotational frequency and wind speed. The aft-most rotor in the plus configuration experienced as large as a 28% decrease in the thrust coefficient when compared to one of the aft rotors in the X configuration for the same hub spacing and flight conditions. Good correlation was found between these wind tunnel tests and flight tests for the front and side rotors in X and plus configurations (7.9–14.2% difference), but a larger difference of 30–41.9% was found for the aft rotors, which is likely due to the different rotor trim conditions.

NOMENCLATURE

A	Rotor disk area (m^2)
C_Q	Torque coefficient, $\frac{Q}{\rho A(\Omega R)^2 R}$
C_T	Thrust coefficient, $\frac{T}{\rho A(\Omega R)^2}$
Q	Motor torque (N-m)
R	Rotor radius (m)
Re_c	Blade tip chord Reynolds number, $\frac{\rho V c}{\mu}$
T	Rotor thrust (N)
V	Free stream velocity (m/s)
$\frac{dF}{dT}$	Force thermal drift rate (N/K)
c	Blade tip chord (m)
α	Vehicle pitch angle (deg)
χ	Wake skew angle, $tan^{-1}\frac{\mu}{\lambda}$
λ	Inflow ratio, $\frac{V sin\alpha + v}{\Omega R}$
μ	Advance ratio, $\frac{V cos\alpha}{\Omega R}$
v	Induced velocity, $\frac{v_h^2}{\sqrt{(V cos\alpha)^2+(V sin\alpha+v)^2}}$
v_h	Hover induced velocity, $\sqrt{\frac{T}{2\rho A}}$
Ω	Propeller rotational speed (rad/s)
ψ	Rotor azimuth angle (deg)
ρ	Air density (kg/m^3)

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*Main author

INTRODUCTION

In recent years, electric vertical takeoff and landing (eVTOL) vehicles have garnered significant interest because of their versatility, mechanical simplicity, and relatively low cost. Many of these vehicles utilize multi-rotor configurations such as coaxial, tandem, and quad-rotors due to their small size and low weight, which allows for rapid changes in thrust by changing the rotational speed of the rotors. This is simpler and easier than changing the pitch angles of the blades via a more complicated swashplate mechanism. These vehicles offer solutions to problems faced by both civilian and military applications, which has caused increased research activity in this field. However, there are multiple opportunities for improvement of these vehicles. One such area is the aerodynamic performance which has implications on the power consumption, battery life, and therefore, on the types and lengths of missions that can be undertaken. To be able to improve the aerodynamic performance, it is necessary to understand the aerodynamic interactions between the rotors for different configurations and flight regimes. In addition, it is important that the behavior of the rotors be examined in low Reynolds number regimes, addressing the flight conditions of many commercially available eVTOL vehicles (e.g., surveillance or recreational drones) that have small rotors and typically fly at velocities below 20 m/s.

Related work has been done by NASA using the Multirotor Test Bed (MTB), which can employ up to six rotors. The results presented by Russell and Conley (Ref. 1) showed the effects of changes in the pitch angle of the vehicle on the thrust produced by each one of the rotors. The spacings between the rotor hubs were also varied. In addition, they compared these thrust values to an isolated rotor.

Zhou et al. (Ref. 2) focused on the evaluation of aerodynamic interactions in a side-by-side rotor as a function of the hub spacing. That work, however, examined the effects of the rotors in axial flow, which is not comparable to multi-rotor vehicles that are operating in edgewise flight. Shukla and Komerath published several works that experimentally investigated the aerodynamic interactions of small UAV configurations. In one such work (Ref. 3), a bi-rotor configuration was evaluated and vortex–vortex, vortex–vortex sheet, and vortex–hub interactions were investigated. Two more works, (Refs. 4,5), investigated the interactions for coaxial rotors and side by side rotors, respectively.

Although there is limited research in the experimental evaluation of the aerodynamic effects between edgewise rotors, there has been a plethora of research done in the computational evaluation of these effects. Ventura-Diaz and Yoon (Ref. 6) investigated the effects of the position and type of the rotor mount on the performance metrics of multi-rotor UAVs. In addition, the authors performed computational analyses to address the changes in the vehicle performance when the number of rotors change. However, the effects of the hub spacings were not addressed. Yoon et al. (Ref. 7) presented the computational evaluation of the rotor–rotor interactions and the effects of the hub spacing on the average thrust coefficient of the rotors in a quad-rotor configuration in edgewise flight using detached eddy simulations (DES). They found that hub spacings indeed play a significant role in the vertical forces produced by quad-rotor systems in hover. Similarly, Healy et al. (Ref. 8) presented computational results that shed light on some of the phenomena experienced by the rotors and the effects of vertical and horizontal spacing on the overall rotor thrust and torque. Their findings showed that there was a lift reduction and a torque penalty on the aft rotor in comparison to an isolated rotor in the same conditions.

Piccinini et al. (Ref. 9) showed similar results for dual-rotors in side-by-side and tandem configurations. The performed investigation showed that in the side-by-side rotor configuration, there was only a small reduction in the thrust produced by both rotors, whereas a significant loss of thrust of about 40% was observed for the tandem configuration. Healy et al. (Ref. 10) also examined the effects of cant angle on the performance metrics of tandem rotor configurations. They found that in all cases there was a degradation in the lift of the aft rotor in the tandem configuration regardless of the cant angle due to the downwash induced by the wake of the front rotor on the aft rotor. In this paper, all analyses were performed at a single hub spacing and at a defined disk loading.

Although multiple works exist on the computational evaluation of multi-rotor aerodynamic interactions, most of such numerical investigations have not been validated against experimental measurements, simply because of a lack of experimental data available for relevant configurations. Conley et al. (Refs. 11, 12) performed one of such comparisons on the results obtained from the NASA MTB with data obtained from the rotorcraft analysis tools RotCFD and CHARM. They found that the discrepancies that were observed between the computational results and experimental data were smallest for the isolated rotor case. Additionally, it was found that for RotCFD there were higher discrepancies for the quad-rotor cases than for the six-rotor cases, and for CHARM there were higher discrepancies for the dual-rotor configurations than for the quad-rotor configuration. The CHARM analysis also found that increasing the number of rotors increased the difference in rotor performance between the simulations with the body and the simulations without the body. It was found that RotCFD performed better at predicting the performance characteristics than CHARM did with average discrepancies as high as 11.8% between the CHARM simulations and the experimental data. This underscores the importance of obtaining experimental databases for validating mid-fidelity computational analysis tools.

Barcelos et al. (Ref. 13) computed the forces and moments acting on quad-rotor configurations using an advanced potential flow method. The different interactions due to vehicle orientation were analyzed and compared with an isolated rotor. They found that the complex, asymmetrical wakes of the lead rotors interacted with the rear rotors and caused different rotor efficiencies. It was generally discovered that an increase in edgewise free stream flow and a decrease in free stream advance ratio will cause the interference effects to show more variations between configurations.

Ventura-Diaz and Yoon (Ref. 14) used NASA's OVERFLOW high-fidelity CFD solver to study the aerodynamics of multi-rotor unmanned aerial vehicles. Two quad-copters were analyzed (the DJI Phantom 3 and the SUI Endurance). The SUI Endurance is specialized for forward flight and it was found that the front rotors' effects on the aft rotors' performance increased with flight speed and angle of attack.

Alvarez and Ning (Refs. 15, 16) used the viscous vortex particle method (VPM) to capture the aerodynamic interactions present in multi-rotor configurations at both low and high Reynolds numbers. They found that the main cause of the drop in blade loading on the aft rotors was the increased induced upwash between the rotors caused by the rotor–rotor interactions. They found that for all advance ratios and Reynolds numbers, the aerodynamic interactions lowered the efficiency of the rotors as the distance between the rotors decreased. Additionally, it was found that the performance drop increased as advance ratio decreased, and that a higher Reynolds number caused the interactions between lightly loaded blades to increase.

Lee and Lee (Ref. 17) investigated the influence of the spacing between rotor tips in UAVs on thrust, wake geometry, flow field, and acoustics using a nonlinear vortex lattice method coupled with a VPM. They found that the tip vortices generated an upwash flow that hindered the downstream convection of the wake and increased wake interactions. This changed the wake structure from a helical form to a more aperiodic structure that caused the tip vortex trajectory to deteriorate more quickly and transition to a less structured, more turbulent wake. This upwash also caused a low-pressure region between the rotors.

The present paper discusses an experimental investigation that aims to improve the understanding of the aerodynamic interactions between the rotors in a multi-rotor vehicle, and to quantify the performance modifications of these rotors arising from such interactions. To this end, hub load measurements were performed on all rotors in different configurations, and flow visualization was conducted on a tandem rotor configuration. The experimental investigation mainly focused on dual- and quad-rotor configurations operating in multiple different forward flight conditions tested in a wind tunnel.

DESCRIPTION OF EXPERIMENTS

Facility

The experiments were conducted in the John J. Harper Memorial Wind Tunnel at the Georgia Institute of Technology (GT). The wind tunnel is subsonic with a closed circuit and is fitted with a breather to allow for atmospheric pressure conditions in the test section. This wind tunnel has a test section that measures 7x9 ft, which allows for large test articles and full vehicle models to be tested. More importantly, for current tests it allows for a large distance between the rotors being tested and the walls of the wind tunnel, minimizing facility effects such as ground or wall effects. For the performed measurements, all rotors had a minimum distance of $6R$ from all walls.

Test Rig

To allow for the evaluation of the aerodynamic interactions between the rotors in a multi-rotor vehicle, a test rig was designed and built. The developed test rig was inspired by the Multirotor Test Bed (MTB) developed by NASA (Ref. 1). The rig was designed and constructed with variability and versatility as the highest priority to enable tests in multiple different configurations. To this end, the test rig has the capability to test one to four rotors with the ability to independently change the lateral, longitudinal, and vertical spacing between the rotors. Additionally, both the overall test rig pitch angle and the relative flow angle (i.e., yaw angle) can be varied. This includes the "plus" and "X" configurations shown in Figure 1, which are the most common quad-rotor orientations when in operation. The test rig has the capability to support hub spacings of up to $3R$ both laterally and longitudinally in the X configuration, and $3R$ to $4.5R$ in the plus configuration. The rig also has a vertical height range of approximately $4.5R$ to $8R$ above the base for each rotor. The minimum hub spacing in the plus configuration was constrained by safety of operation, as any hub spacing smaller than $3R$ would lead to blade strike between rotors.

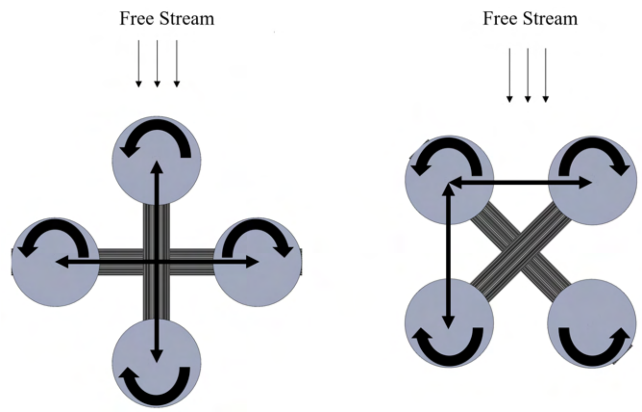


Figure 1. Schematic showing plus and x configurations, rotational directions, and how hub spacing was measured for these configurations

Test Hardware and Instrumentation

Most of the hardware was selected to ensure comparability of the wind tunnel tests with the flight tests performed by research collaborators at the Ohio State University (OSU). Therefore, the motors and rotors used were identical to those on the flight test drone. The motors used were the T-Motor U7-V2.0 KV420. The propellers were the DJI S1000 1552 with a radius of 7.5in. These propellers have two blades that are connected to the central hub of the motor with a hub adapter. Due to the presence of this hub adapter, the root pin of the blade is 0.925 in offset from the center of the hub. Table 1 presents some features of the rotor geometry. These blades have complex geometries and the blade parameters presented in Table 1 are approximations.

Table 1. Rotor specifications

Radius (R)	7.5 in
Rotor disk area	176.71 in^2
Root cutout	1.95 in
Blade tip chord	0.5625 in
Vehicle pitch angle	10°
Number of blades	2
Blade taper ratio	2.44:1
Twist rate	16.7°/R
Hub type	Rigid and hingeless

The instrumentation used to obtain the hub loads data from the individual rotors can be seen in Figure 2. A small hall effect sensor attached to the motor mounting plate was used to measure the rotational speed of the motor with the internal motor magnetic poles. The rotational speed readout from the hall effect sensor was then sent to a computer running an Arduino code that uses a feedback loop that adjusts the RPM of the motor until the desired RPM was achieved. This was the only means of controlling the rotor, i.e., there was no collective pitch or other such control.

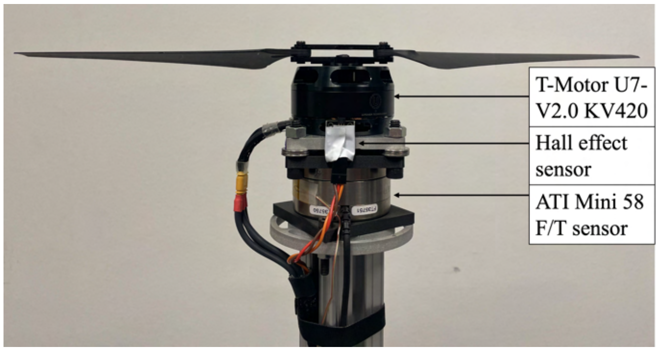


Figure 2. Rotor instrumentation

The loads data were obtained using the ATI mini 58 six-axis force/torque sensors. One sensor was attached to each motor and was able to measure the individual rotor forces and moments in all three axes. The rotor dynamic loads were measured in all six axes and in the post processing procedure, the in-plane forces were disregarded in the experiments as only the out-of-plane forces and torques were of interest.

The raw voltages from the force/torque sensors were amplified using interface power supply (IFPS) boxes and then sent to the computer through the LabJack T7 Pro data acquisition system. These voltages were then converted into forces in MATLAB via each device's specific calibration matrix.

Flow Visualization

In order to visualize the flow structures and wake interactions in the tandem configuration, two-dimensional flow visualization was used. The flow visualization setup is shown in Figure 3. A Photonics Industries dual-cavity Nd:YLF laser with a pulse energy of 30 mJ was used to produce a laser light sheet, which was placed in parallel with, but two inches off the rotational axis of the rotors for reasons of optical accessibility. The laser source was placed outside of the wind tunnel test section and the laser beam was channeled into the test section via a focus lens and two 90° mirrors. The laser beam was then passed through a diffractor which formed the laser sheet. Two Phantom V341 4-megapixel CMOS cameras were oriented perpendicular to the laser sheet to obtain a double-frame image per camera. The upstream camera was centered just aft of the front rotor and the second camera was placed 12 inches downstream of the upstream camera to capture the interactions between the wakes of the two rotors. Each of these cameras captured the flow structures in the plane of the laser light sheet. Seeding particles were evenly dispersed into the test section of the wind tunnel through a pipe with evenly distributed holes upstream of the tandem rotor posts. Using the DaVis PIV software and a programmable timing unit (PTU), 100 images per camera were taken at each test condition to ensure adequate wake progression. These images were taken at a frequency of 402 Hz.

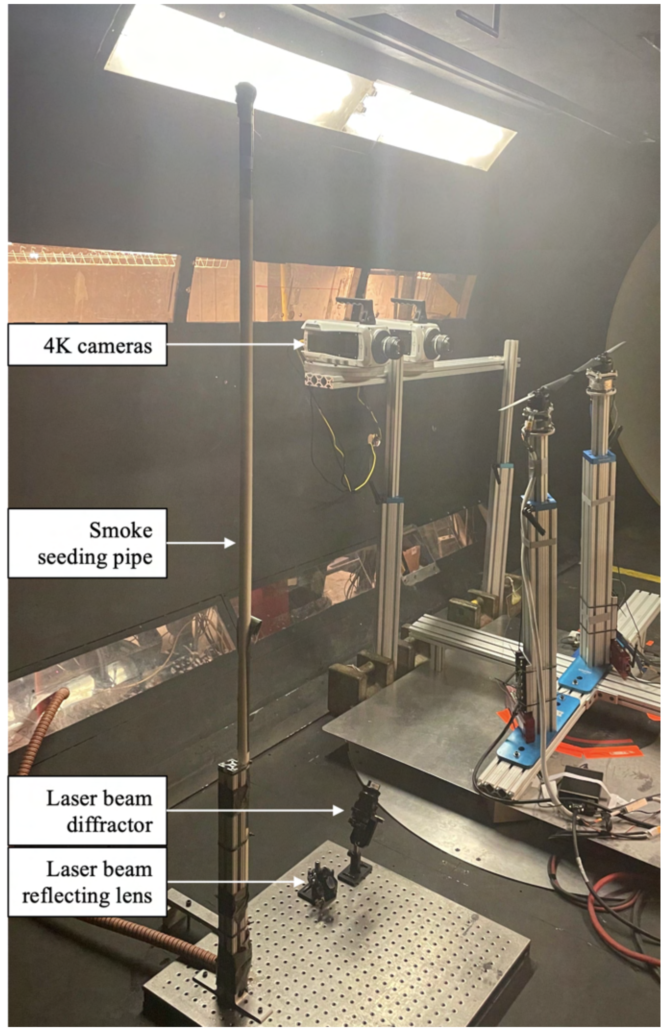


Figure 3. Flow visualization setup for tandem configuration. Wind flows from left to right.

Thermal Drift Mitigation

It is known that silicon based strain gauges are prone to changes in force readings as the temperature changes. As these experiments are performed in a wind tunnel, the changes in the temperature of the force/torque sensors are difficult to predict due to the motor heating and the cooling from the free stream. This means that there must be rigorous methods put in place to ensure that the change in loads due to the changes in temperature are mitigated.

In the design of the test rig, multiple steps were taken to minimize the effects of temperature changes on the force/torque sensors. Figure 4 shows these design features. First, the test rig was designed such that there was a physical gap between the motor and the force/torque sensor for insulation purposes. In addition, the mounting plate between the force/torque sensor and the stand and the mounting plate between the force/torque sensor and the motor were made from 3-D printed carbon fiber infused nylon because of its low thermal conductivity and expansion. The force/torque sensors were fitted with a reflective aluminum insulation thermal shield that blocks direct contact between the free stream from the wind tunnel and the force/torque sensors.

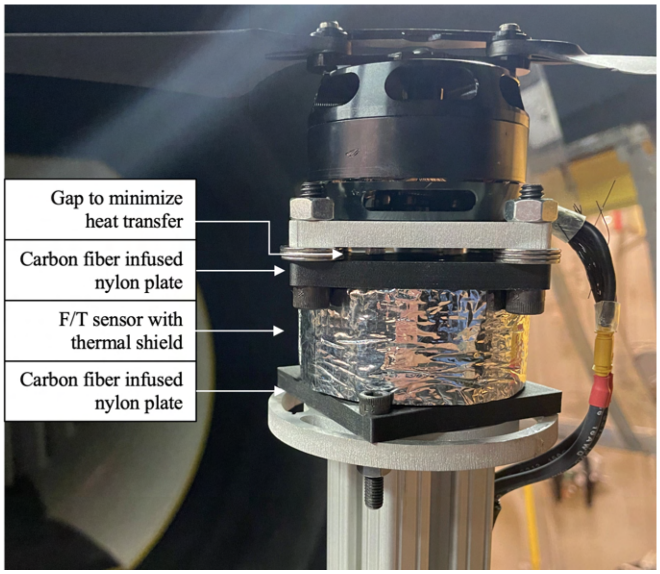


Figure 4. Motor and F/T sensor setup for thermal drift mitigation

Although these preventative measures decrease the thermal drift, they do not eradicate it completely. Thus, each one of the force/torque sensors was fitted with K-type surface thermocouples to obtain the real-time temperature data to be used in the compensation for the changes in the temperature as the data collection proceeded. The data collection procedure itself also contained some level of thermal drift mitigation. Before each data set was taken, a bias data set was collected with the motors off and zero free stream velocity. This data set represented the residual loads of the force/torque sensors which were then subtracted from the data set taken with the motors running.

Test Matrix

Table 2 shows the operating conditions that were tested. Some of these operating conditions were decided upon to allow for comparability with the flight tests performed by research partners at the Ohio State University (OSU). In particular, the rotor hub spacings of 2.6R and 3.63R correspond to the dimensions of the vehicle used at OSU. The rotor RPMs were varied from 4500 RPM to 6000 RPM in increments of 500 RPM to allow for a range of test conditions while matching specific rotational speeds observed during the flight tests.

The spacing between the rotors in the multi-rotor test rig was varied. The rotor hub spacings of 2.6R in the X configuration and 3.63R in the plus configuration match with the flight test drone. This was done in order to accurately mimic the potential aerodynamic interactions that occur during flight tests. The other values for the spacing between rotors were chosen to allow for an evaluation of the effects of the spacing between rotors on the aerodynamic interactions. The overall test rig pitch angle representing vehicle pitch was set to 10° nose down. This parameter was also determined based on the attitude of the vehicle in forward flight during the flight tests.

The effects of the aerodynamic interactions on the rotor performance of each individual rotor were studied in different rotor configurations. The primary configurations studied were the quad-rotor in the plus and X configurations as shown in Figures 5(b) and 5(a). These configurations represent some of the most common quad-rotor orientations. Hence, the results from these tests provide valuable information for the types of interactions that occur on such vehicles. In addition to these configurations, a tandem rotor configuration was also examined. This simpler configuration also presents valuable insight into the effects of the wake and vortex interactions between the forward rotor and the aft rotor including rotor performance effects.

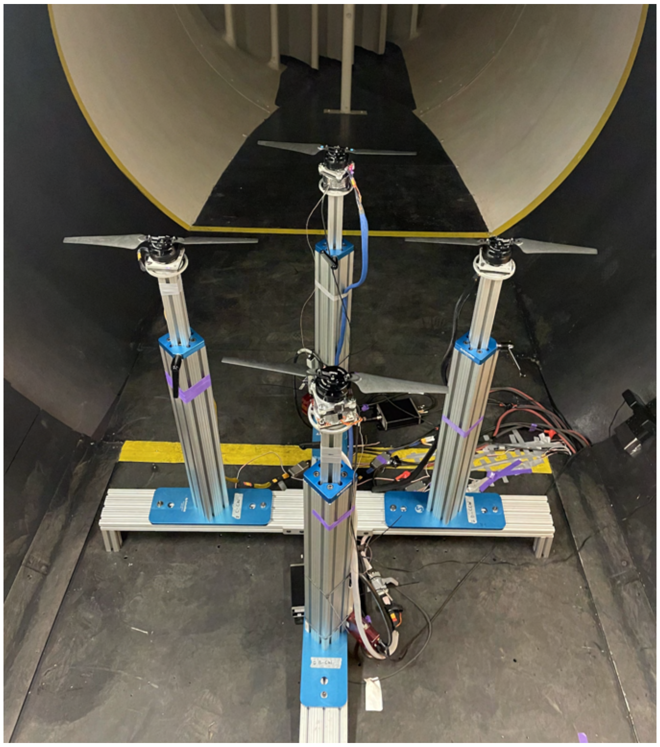
Table 2. Operating conditions

Propeller RPM	4500, 5000, 5500, 6000
Free stream velocity (m/s)	8, 10,12
Min tip chord Reynolds number	73,000
Max tip chord Reynolds number	118,000
Rotor hub spacings	2.1R, 2.5R, 2.6R, 3R, 3.5R, 3.63R

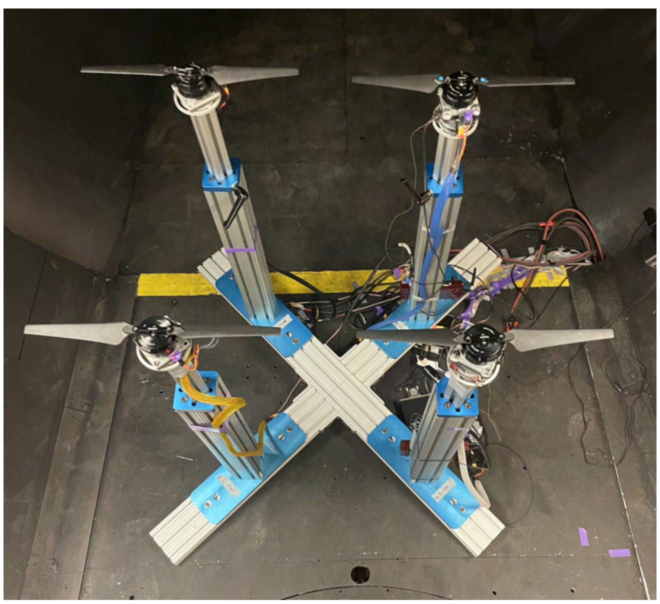
DATA COLLECTION AND POST PROCESSING

Data Collection Procedure

The data collection procedure for each configuration and flight condition started with collecting a baseline data set. This data set was collected with the wind tunnel off and the motors off. This was done to measure the residual loads on the force/torque sensors from the motor and mounting plate connections. Next, the wind was turned on until the desired test section speed was achieved. Once this occurred, the motors were switched on until the desired motor rotational speed was achieved. Finally, the raw voltages and real-time temperature of the force/torque sensors were collected. For each of the force/torque sensors, 200 data points were acquired. This translated to about a minute of data collection for each operating condition. A convergence study was performed on the number of data points to confirm that the loads were converged to the correct value and that they were not biased by too low of a sample size. 200 data points gave the same level of accuracy as with obtaining more data points, and provided a good compromise between the accuracy and the testing time needed to get through the large test matrix.



(a) Plus configuration



(b) X configuration

Figure 5. Test rig setup with four rotors in the plus and X configurations installed in the Harper Wind Tunnel at Georgia Tech

Post Processing and Thermal Compensation

Once all of the raw voltages and temperatures from the force/torque senors were obtained, the baseline voltages collected with the wind and motors off were subtracted from the powered data set to remove the residual voltages. The raw voltages were then converted into forces and moments using the calibration matrices for each one of the force/torque sensors.

Once the forces are obtained, the effects of the changes in the temperature of the force/torque sensors on the obtained forces need to be accounted for. This was done by plotting the force obtained from the force/torque sensor against the changes in the temperature of the sensor without external load on the sensor. A linear curve fit was obtained, and the slope of this line represents the change in the force obtained from the sensor per Kelvin change in the temperature of the sensor. An example of such a plot and trend line is shown in Figure 6, where it can be seen that the force changed by 1.67 N per Kelvin.

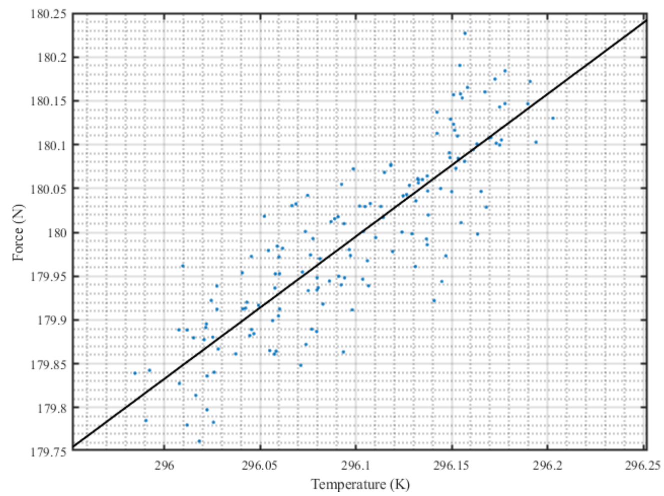


Figure 6. Thermal compensation trendline

From Figure 6, it is seen that as the temperature of the force/torque sensor decreased, the obtained force also decreased without additional external load added. During tests, the observed temperatures of the force/torque sensors decreased, which indicated that the forces that were measured while the test progressed were lower than the actual (i.e real) values, and thus required mathematical thermal compensation. Equation 1 shows the thermal compensation equation used to account for the temperature changes. A similar process was used by Russel et. al (Ref. 18) and Battey and Russell (Ref. 19).

$$F_{corrected} = F - \frac{dF}{dT}\Delta T \tag{1}$$

Figure 7 shows the results from one such thermal compensation procedure. The figure shows the baseline forces measured at a given temperature (original data set), the data set obtained after the temperature had changed from when the baseline data set had been collected which (drifted data), and the corrected forces using Eq. 1. Figure 7 shows that between the baseline (original) data set and the drifted data set, the forces had changed to about -1.38 N on average without additional external load applied to the force/torque sensors. However, the thermal compensation procedure corrected for this and the average of the corrected loads was at the same level of the baseline (original) loads, as desired.

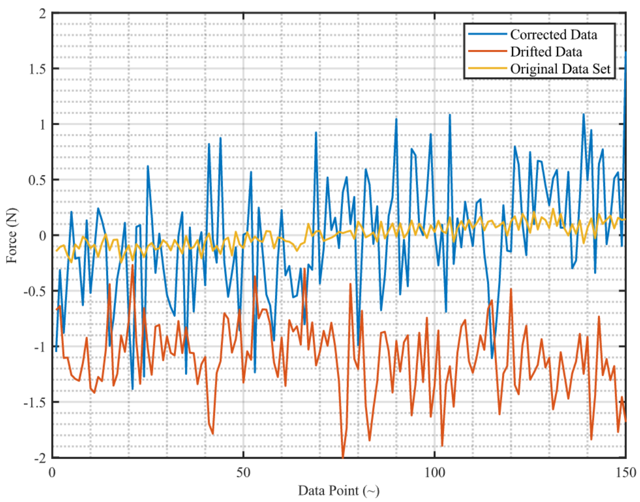


Figure 7. Example thermal drift correction

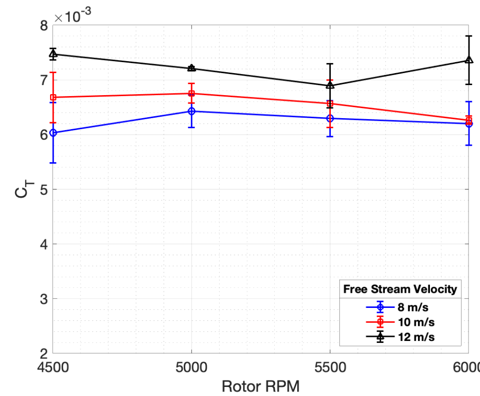
After the thermal compensation was performed on the data sets, an average of the 200 data points was calculated. For each combination of configuration and rotor operating condition, the data acquisition was conducted three separate times to ensure repeatability. An average of the forces and moments obtained from the three tests was calculated, and the 2σ standard deviation of these three tests was obtained and used to determine the uncertainty bounds in the presentation of the results for each test condition.

RESULTS AND DISCUSSION

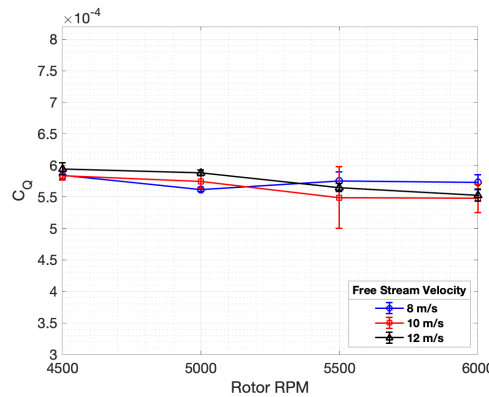
Isolated Rotor

Figure 8(a) presents the thrust coefficient (C_T) of the isolated rotor plotted against the rotational speed of the rotor for all three wind speeds that were examined. This isolated rotor was the aft rotor used in all other configurations running without the presence of any other rotors. The rotor had a 10° shaft tilt angle, which was constant for all configurations and wind speeds. The results show that there was almost no variation in the thrust coefficient as the rotational frequency increased for a given wind speed. This implies that the thrust increased with the square of rotational frequency, which is consistent with the expected result that the rotor produces more thrust at higher rotational speeds due to the increase in tangential velocity across the rotor disk. In addition, it is seen from Figure 8(a) that at a given rotational speed, the thrust coefficient increased as the wind speed increased. These results are consistent with the experimental results obtained by Jamaluddin et al. (Ref. 20) and the numerical analysis by Yang et al. (Ref. 21) performed for an isolated rotor.

Figure 8(b) presents the torque coefficient (C_Q) for the same isolated rotor. In this case, the general trend observed was a slight decrease in the torque coefficient as rotational speed



(a) Thrust coefficient



(b) Torque coefficient

Figure 8. Thrust and torque coefficients of the isolated rotor for various rotational frequencies and free stream velocities

increased, although the changes were very small, and within the uncertainty bounds. These results are due to the fact that there is better airfoil performance at higher Reynolds numbers. These results are consistent with the findins from the MTB experiments performed by Russel and Conley Ref. ((Ref. 1)). Similarly, the wind speed had very little effect on the torque coefficients of the rotor. These isolated rotor results were used as a baseline for comparison of the effects of the aerodynamic interactions on the performance of the rotors in the multi-rotor configurations.

Tandem Configuration

The tandem rotor configuration is relatively simple and provides information about the direct effects of the interactions between the wakes and vortices of both rotors on the performance metrics. The aft rotor measurements are presented in this section and compared to the isolated rotor. In addition to the loads data, flow visualization was also performed on the tandem configuration to observe the generation of flow structures and interactions that the wake and the flow structures have with the rotor blades of the aft rotor. Figures 9(a) through 9(d) show the thrust coefficients for the aft rotor in the tandem configuration. The legend in these plots show three different advance ratios that correspond to wind speeds of 8, 10, and 12 m/s. These wind speeds were set to allow for comparability with flight test data from OSU.

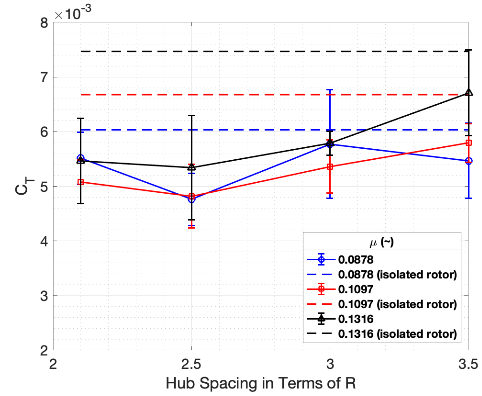
The figures show that there was a reduced thrust coefficient of the aft rotor when compared to the isolated rotor. This shows that the aft rotor was directly affected by the upstream rotor. It is also apparent that the hub spacing played a significant role on the thrust coefficient of the aft rotor. Figures 9(a) through 9(d) show that as the hub spacing decreased, the reduction in the thrust coefficient of the aft rotor was more pronounced when compared to the isolated rotor, i.e., more significant wake interactions occurred. For example, Figure 9(d) shows that at a hub spacing of $3.5R$, the thrust coefficient of the aft rotor was 13.5% lower than that of the isolated rotor for an advance ratio of 0.0987. When the hub spacing was decreased to $2.1R$ the decrease in the thrust was 24.3% for the same advance ratio.

It is also seen from these figures that there were no significant effects from the changes in the wind speed on the thrust coefficient of the aft rotor. This is in stark contrast to what was observed in the isolated rotor case and it could indicate that the rotor–rotor interactions offset the affects of wind speeds. Although consistent trends were seen for the tandem rotor configuration, there were few anomalous data points that did not match the overarching trends. For example, the data for 5500 RPM in Figure 9(c) show deviations from the expected trends that may be attributed to the complexity of the aerodynamic interactions that occurred at this operating condition. A more in-depth investigation is required to examine the physical reason for these discrepancies.

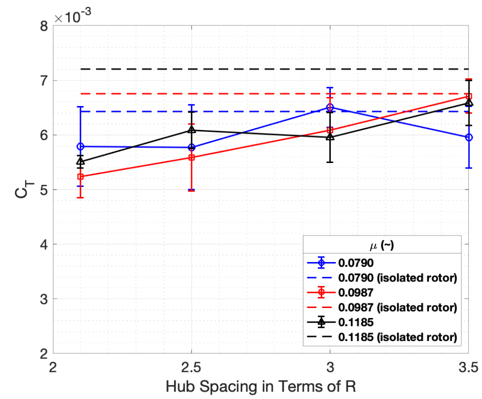
Figures 10(a) through 10(d) present the torque coefficients for the aft rotor. There was little effect of the hub spacing on the torque coefficient of the aft rotor in the tandem configuration, although a slight decrease in the torque coefficient was observed as the hub spacing increased. It is also seen that when compared to the isolated rotor, more torque was required to maintain the rotational speed of the aft rotor in the tandem configuration than in the isolated rotor configuration. For a rotational frequency of 6000 RPM, the required torque was 11.4% to 18.9% higher than what was required in the isolated rotor case for the different advance ratios and hub spacings. The results also do not show much effect of the wind speed on the torque coefficients for the aft rotor, which was similar to what was observed for the isolated rotor.

Tandem Configuration Flow Visualization

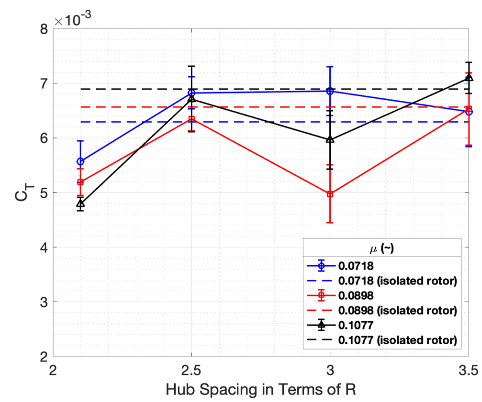
Flow visualization was performed on the front and aft rotors in the tandem configuration to observe the production and convection of flow structures and to gain insights and help interpret some of the loads measurements. This was done using a laser sheet located in the streamwise plane, 2 inches starboard of the plane passing through the two rotor hubs. Figure 11 shows a schematic of the flow visualization images and the direction of the free stream. Figures 12(a) through 12(f) show the images that were captured for the tandem rotor at 4500



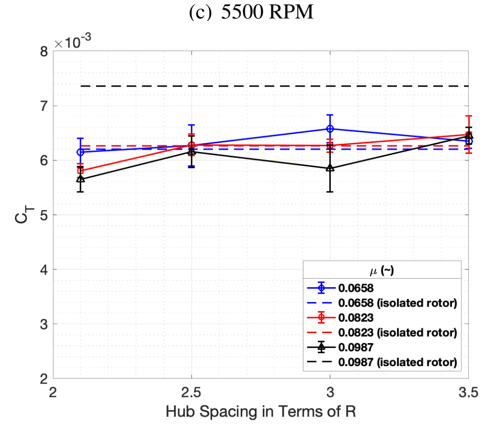
(a) 4500 RPM



(b) 5000 RPM

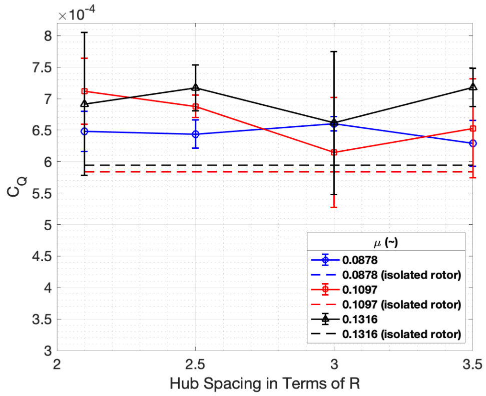


(c) 5500 RPM

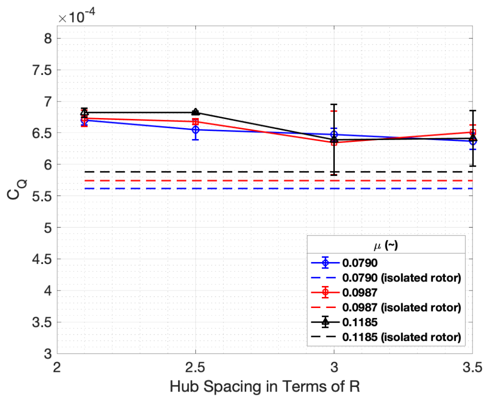


(d) 6000 RPM

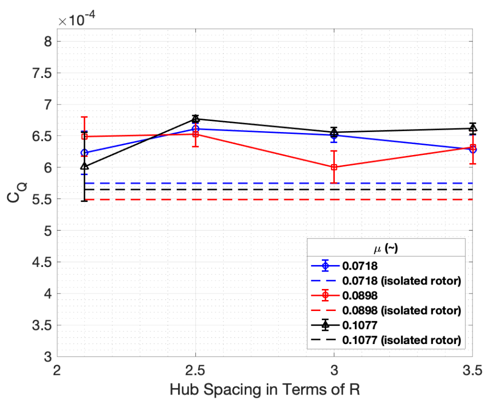
Figure 9. Thrust coefficient for the aft rotor in the tandem configuration for various hub spacings, advance ratios, and rotational frequencies



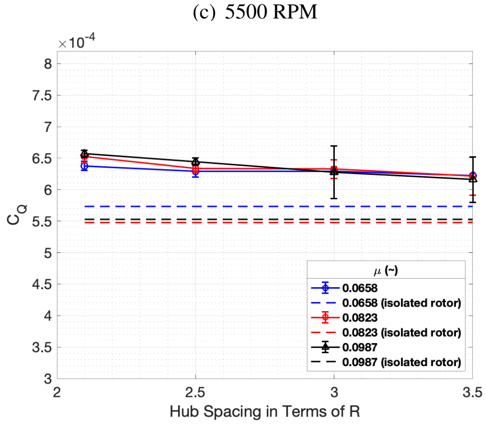
(a) 4500 RPM



(b) 5000 RPM



(c) 5500 RPM



(d) 6000 RPM

Figure 10. Torque coefficient for the aft rotor in the tandem configuration for various hub spacings, advance ratios, and rotational frequencies

RPM, a hub spacing of $2.1R$, and two wind speeds: 8 m/s and 12 m/s. These figures also show three contiguous frames for the operating conditions presented to show the propagation of wake structures as time progressed. It is seen from these images that the wake of the front rotor was not directly sucked into the aft rotor because the aft rotor was higher up than the front rotor due to the test rig pitch angle relative to the free stream. Therefore, there was no direct impingement of the wake of the front rotor on the aft rotor.

These images show the direct effect of the wind speed on the interactions that occur between the flow structures. The 8 m/s wind speed case is presented in Figures 12(a) through 12(c), and young tip vortices were seen to have been generated in Figure 12(a). In Figure 12(b), 67.2° of rotor azimuth later, these tip vortices had aged and were convected down and aft from their original locations. The images show that although the tip vortices were in close proximity, these vortices had minimal interactions when compared to the 12 m/s case (shown in Figure 12(d) through Figure 12(f)). In Figure 12(d), the young vortices are seen shortly after they were produced at the blades. The image shows that these vortices intersected with each other and with older vortices that were generated during previous revolutions. These interactions are also shown in Figure 12(f), which is 134.4° of rotor azimuth after Figure 12(d). These interactions of the flow structures lead to induced velocities that, in turn, induce forces and moments that affect the performance and loads of the rotors.

For a rotor in forward flight, the rotor wake and blade tip vortices are convected downstream by the component of the free stream velocity that acts parallel to the rotor disk (μ) and downward by the component that is normal to the disk (λ). Using momentum theory, the wake skew angle (χ) can be calculated. These calculations were done using the equations presented in (Ref. 22) and also listed in the nomenclature. The advance ratio and inflow ratio are dependent upon the measured rotor thrust and wind speed (i.e., the simulated forward flight speed).

In addition to the values calculated from momentum theory, the wake skew angles for the front and aft rotors were estimated from PIV images based on the locations of the rotor blade tip vortices as they convected downstream. These values were obtained using the Q-criterion, or the difference between the vorticity rate tensor and the strain rate tensor. When Q is greater than zero, it indicates the presence of a vortex. As such, a higher Q generally means a stronger vortex region. The vortex region was identified, and the centroid of the nonzero Q values (and therefore the vortex) was located. This approximate location of the vortex was located for up to 100 images, giving a collection of vortex locations. Then, a linear fit was done on the collection of vortex centroid coordinates, which was used as an estimate for the vortex trajectory. The angle between the vortex trajectory vector and the rotor shaft axis was found. This corresponded to the value of the wake skew angle.

It was expected that the location of the aft rotor relative to the front rotor had no effect on the wake skew angle of the front rotor. Therefore, Table 3 presents the wake skew angles of the front rotor at different advance ratios but for only one hub spacing of 3.5R. This table presents the measured values from the flow visualization images. For all rotational speeds examined, the table shows that the wake skew angle increased as the wind speed increased, which was expected. It is also expected that the skew angle decreases with increasing rotational speed (higher rotor thrust and thus higher downwash velocities). This trend was observed for all of the RPMs and wind speeds, with one outlier being the operating condition of 5500RPM at the lowest wind speed of 8 m/s.

For the aft rotor, it is important to include the effects of the hub spacing on the wake skew angle in addition to the effects of the advance ratio, because this variation may lead to different skew angles indicating different levels of wake interactions between the front and aft rotors. These skew angles for the aft rotor are presented in Table 4.

Table 3. Front rotor wake skew angles

3.5R					
Rotor RPM		4500	5000	5500	6000
Wind Speed (m/s)	8	23°	21°	25°	18°
	10	31°	27°	25°	23°
	12	39°	35°	32°	28°

Table 4. Aft rotor wake skew angles

2.1R					
Rotor RPM		4500	5000	5500	6000
Wind Speed (m/s)	8	64°	63°	61°	60°
	10	65°	64°	63°	63°
	12	63°	65°	64°	64°
2.5R					
Rotor RPM		4500	5000	5500	6000
Wind Speed (m/s)	8	64°	61°	59°	54°
	10	67°	65°	63°	68°
	12	67°	67°	66°	66°
3R					
Rotor RPM		4500	5000	5500	6000
Wind Speed (m/s)	8	67°	65°	61°	58°
	10	69°	67°	66°	63°
	12	70°	69°	70°	73°
3.5R					
Rotor RPM		4500	5000	5500	6000
Wind Speed (m/s)	8	68°	64°	67°	63°
	10	69°	70°	69°	69°
	12	72°	71°	71°	74°

From Table 4, it is seen that the spacing between the rotors had a significant effect on the wake skew angles of the aft rotor. Generally, it is seen that as the hub spacing increased, the wake skew angle of the aft rotor also increased for a given advance ratio. When the aft rotor was in close proximity to the front rotor, the downwash and the flow structures from the front rotor affected the aft rotor and caused the wake of the aft rotor to also convect downward more rapidly, essentially increasing the effect of the downward velocity component. Therefore, the increase in wind speed had less of an effect on the wake skew angles at lower hub spacings, because the wake effects from the front rotor offset the effects from the increase in streamwise velocity.

The wake skew angles for the aft rotor were also significantly different from the angles obtained for the front rotor. Besides the wake effects, the skew angles reported for the front rotor were estimated based on the vortices that originated from the aft portion of the rotor, and those reported for the aft rotor were estimated with the vortices that originated from the front portion of the rotor. This was decided based upon the field of view of the setup used for the flow visualization, but it increases the uncertainties in determining the wake skew angles. The measured values for the wake skew angles were also compared to the theoretical values obtained based on momentum theory. The differences between the measured and calculated wake skew angles were below 15%, which gives confidence in the accuracy of the estimates from the optical flow measurements.

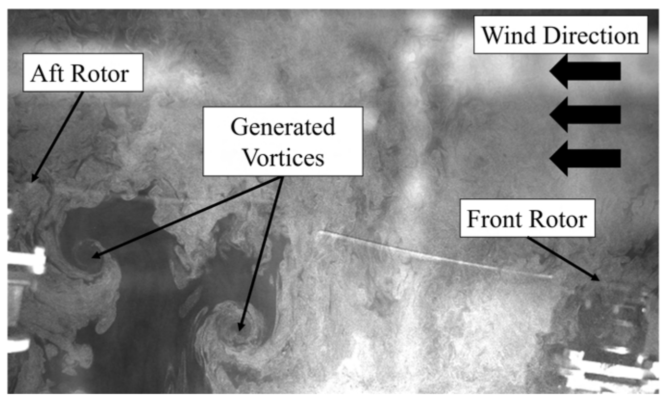
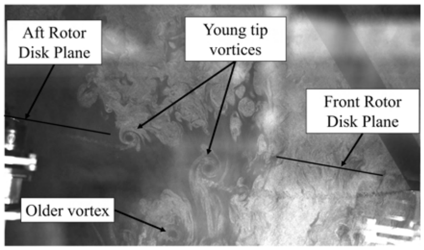


Figure 11. Flow visualization schematic

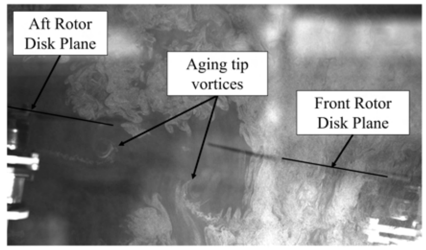
Tandem Configuration PIV

A limited number of PIV measurements were performed and the time average of 100 images calculated (maximum available number of images) in order to gain insight into the interaction between the front and aft rotors in the tandem configuration. The free stream velocity was subtracted from the velocity vectors to isolate the effects from the rotors. Figure 13 shows the velocity component normal to the rotor disk measured 0.05R above the rotor disk for an advance ratio of 0.1316 at hub spacings of 2.1R and 3.5R, respectively. In Figure 13(a), where the hub spacing was largest (3.5R), the aft rotor blade was located far downstream beyond the field of view of the camera. Despite its distance from the visualized region of interest, it can be observed that there was a significant downward velocity normal to the rotor disk closer to the hub of the aft rotor.

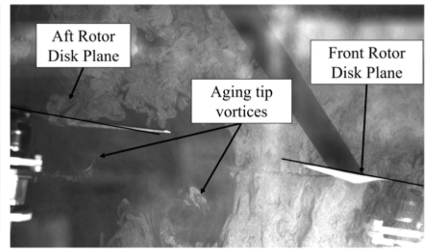
This negative normal velocity (i.e., downwash) suggests the presence of induced inflow effects from the front rotor. Fig-



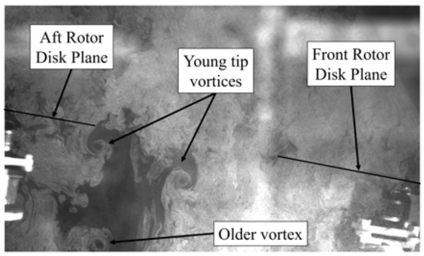
(a) V=8m/s, Frame 1



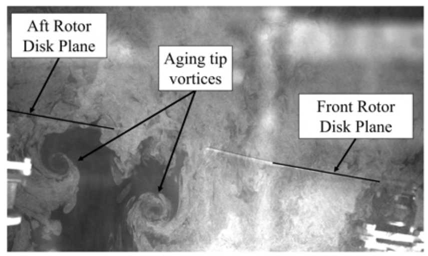
(b) V=8m/s, Frame 2



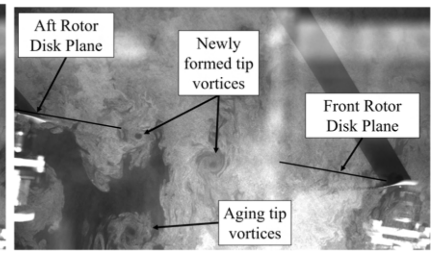
(c) V=8m/s, Frame 3



(d) V=12m/s, Frame 1



(e) V=12m/s, Frame 2



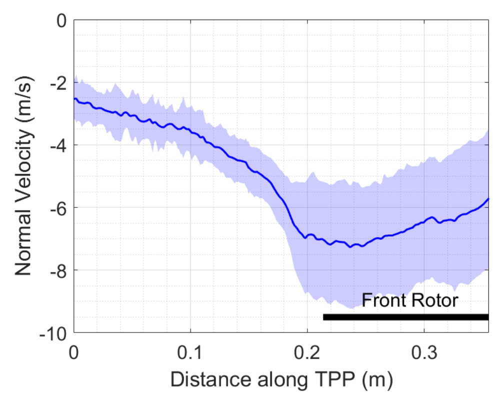
(f) V=12m/s, Frame 3

Figure 12. Tandem Rotor flow visualization images for 2.1R hub spacing, 4500 RPM, $\Delta\psi$=67.2° between frames

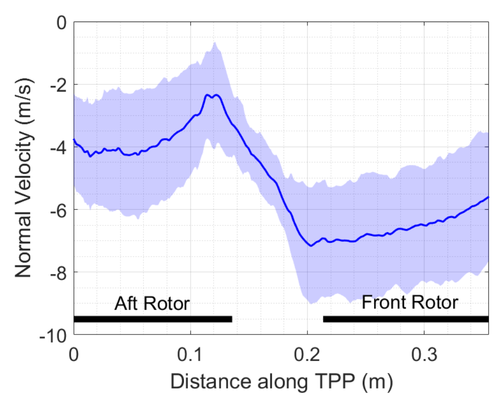
ure 13(b) shows that for smaller hub spacings (here, $2.1R$) the aft rotor blades were located within the region of increased inflow/downwash from the front rotor (compare Figure 13(a)). For these blades, the higher magnitude of inflow (or negative normal velocity, as shown in these graphs) reduces the effective angle of attack at the blades. Consequently, less thrust is generated at the aft rotor, which was consistently seen throughout the investigations. Furthermore, as the spacing between the rotor hubs increases, the magnitude of the downward normal velocity that the aft rotor sees decreases due to dissipation of the front rotor wake and hence reduced interactions. This results in an increase in thrust with increasing hub spacing, as indicated by the trend observed in Figure 9(a) for higher advance ratio cases. For the lower advance ratios, the unsteady interaction of the aft rotor with its own wake and less wake interactions with the front wake due to lower wake skew angles (as previously discussed) may offset the induced inflow effect.

Quad-rotor Plus Configuration

The next configuration that was examined was the quad-rotor plus configuration; see Figures 1 and 5. Similar to the tandem rotor configuration, only the results of the aft-most rotor are presented. Figures 14(a) through 14(d) present the thrust coefficients for the aft rotor in the plus configuration. These figures show that there was a reduction in the thrust coefficient of the aft rotor as the hub spacing decreased, as was expected. In this configuration, the presence of the front and side rotors cause significant changes in the dynamics of the flow upstream of the aft rotor which lead to the degradation in the performance. As the hub spacing between these four rotors increases, the distance between the forward and aft rotors increases, and the side rotors move further out. This means



(a) Hub spacing of 3.5R



(b) Hub spacing of 2.1R

Figure 13. Velocity normal to the rotor disk measured at $0.05R$ above the rotor disk for an advance ratio of 0.1316 and a rotational frequency of 4500 RPM (shaded area indicates ± 1 standard deviation)

that the effect of the wakes of all three rotors on the aft rotor decreases. As seen from Figure 14, there was a clear thrust deficit in the aft rotor when compared to the isolated rotor. It is seen that the thrust of this rotor was almost recovered far the larger hub spacing. For example, Figure 14(b) shows that at the closest hub spacing of $3R$, there was a 15.3% decrease from the isolated rotor case for an advance ratio of 0.1185, but it was only 4.2% at $3.63R$.

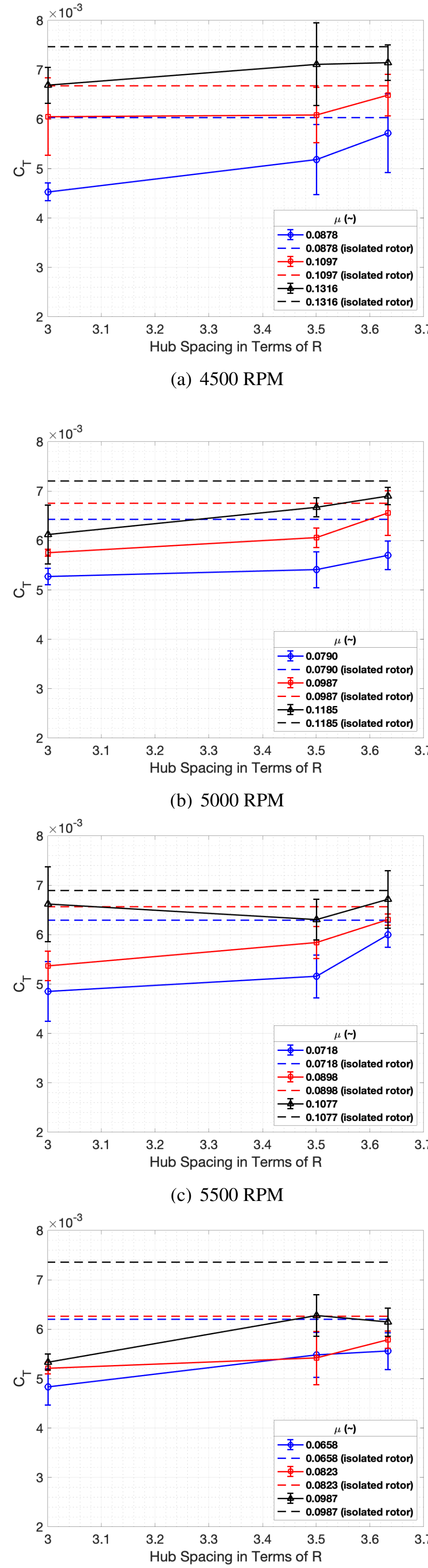
In this configuration, there was also a clear difference in the thrust coefficient of the aft rotor as the advance ratio changed. For all the rotational speeds examined, there was an increase in the thrust coefficient as the advance ratio increased for the aft rotor. In general, is known that the wake skew angle is larger at higher advance ratios a higher advance ratio which means that the wake is blown downstream more rapidly, leading to less interactions between the front and the aft rotors. Further exploration into this will be undertaken with more elaborate PIV measurements to understand the physical reason for this phenomenon.

Figures 15(a) through 15(d) present the torque coefficients for the aft rotor in the plus configuration. Similar to the observed trend in the tandem rotor configuration, the results show that there was an overall decrease in the torque coefficient as the hub spacing increased. For example, at a rotor RPM of 5000 and an advance ratio of 0.079 there was a decrease in the torque coefficient by 5.5% when the hub spacing was increased from $3R$ to $3.63R$. This is the expected trend as the influence of the front and side rotors increases as the distance between the rotors is reduced, and it can be seen in Figure 15(b). However, unlike the tandem configuration, it was observed that the necessary torques were of similar magnitudes as those of the isolated rotor case. It is also seen that there is little effect of the wind speed on the required motor torque to maintain a given rotational speed.

Quad-rotor X Configuration

The final investigation was done on the quad-rotor X configuration. In this configuration, there were two aft rotors which are subjected to aerodynamic interactions from both of the front rotors and from each other. The results are presented for only one of the aft rotors, as the configuration is symmetric. Figures 16(a) through 16(d) present the thrust coefficients for the left aft rotor. There was a significant reduction in the thrust of the aft rotors as the hub spacing decreased. Figure 16(b) shows that the decrease in the thrust coefficient relative to the isolated case increased from 15.3% to 18.3% when the hub spacing decreased from $3R$ to $2.1R$ at an advance ratio of 0.1185. This outcome indicated detrimental aerodynamic interactions on the aft rotor. As the hub spacing increased, the front rotor was less influential on the performance of the aft rotor and, therefore, the performance of the aft rotor improved. In this configuration, it was also seen that the wind speed played a minimal role in the changes of the thrust coefficients for the different rotational speeds that were examined.

Figures 17(a) through 17(d) show the torque coefficients for the left aft rotor. There was very little change in the torque



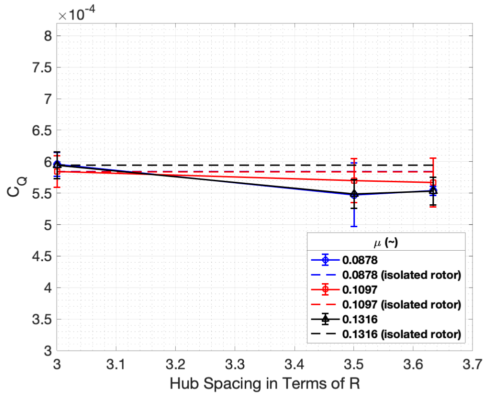
(a) 4500 RPM

(b) 5000 RPM

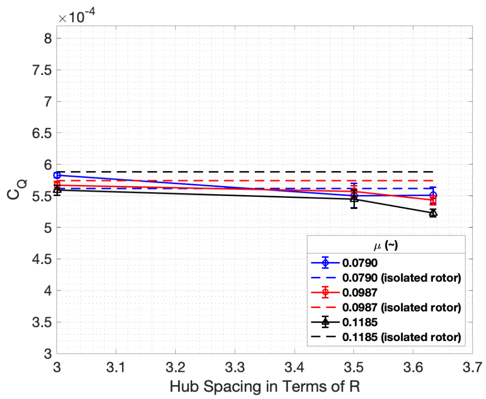
(c) 5500 RPM

(d) 6000 RPM

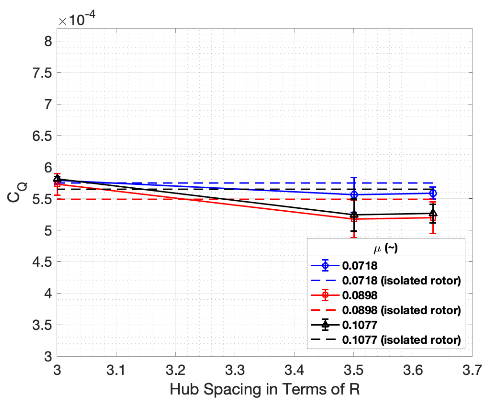
Figure 14. Thrust coefficient for the aft rotor in the plus configuration for various hub spacings, advance ratios, and rotational frequencies



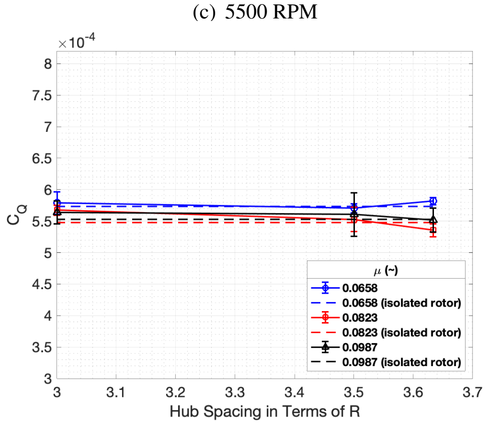
(a) 4500 RPM



(b) 5000 RPM

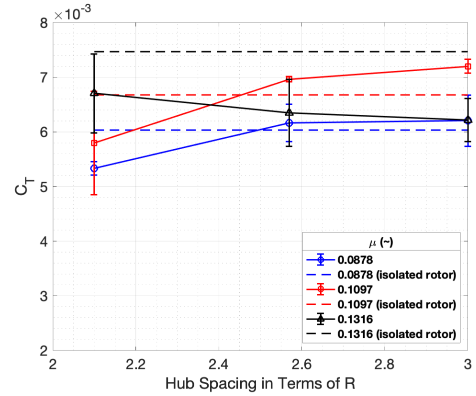


(c) 5500 RPM

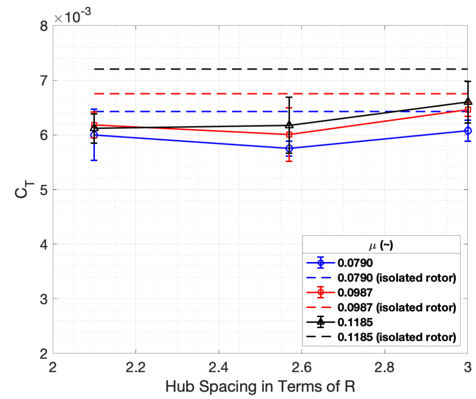


(d) 6000 RPM

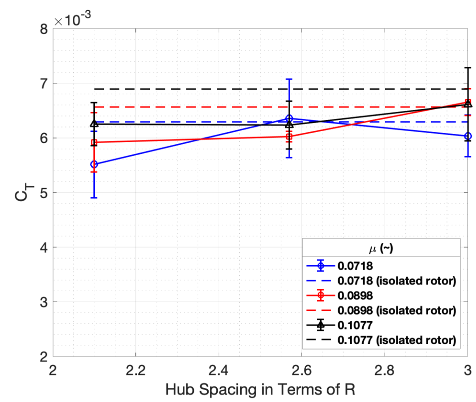
Figure 15. Torque coefficient for the aft rotor in the plus configuration for various hub spacings, advance ratios, and rotational frequencies



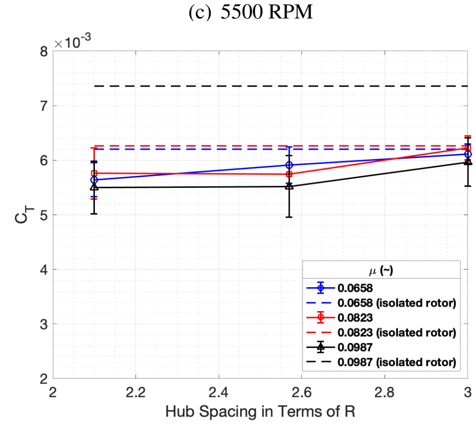
(a) 4500 RPM



(b) 5000 RPM



(c) 5500 RPM



(d) 6000 RPM

Figure 16. Thrust coefficient for the left aft rotor in the X configuration for various hub spacings, advance ratios, and rotational frequencies

coefficient of the aft rotors from changes in hub spacing. It can also be seen that the change in wind speed did not cause significant changes in the measured torque values. However, the required torque was higher for the X configuration than for the isolated rotor case. Figure 17(c) shows that at an advance ratio of 0.0898, the required torque coefficient was between 9.8% and 12.6% higher than for the isolated rotor case.

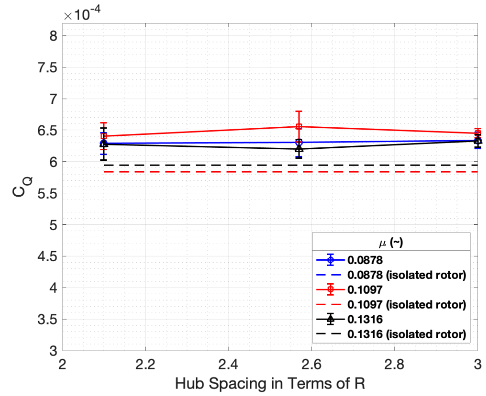
Variations Between Configurations

Examining the thrust coefficients of the plus configuration in comparison to the tandem configuration, it can be seen that the addition of the two side rotors caused a larger thrust deficit on the aft rotors. This became particularly evident as the rotor RPM increased and the hub spacing decreased, resulting in more intense interactions between the three upstream rotors and the measured aft rotor; see, e.g., Figures 9(d) and 14(d). The same trend was observed in the X configuration, and it was expected as the addition of more rotors introduces more aerodynamic interactions. These results are similar to those obtained from the numerical analysis performed by Hwang et al. (Ref. 23), although the computations were performed to include a fuselage in the multi-rotor vehicle which may cause modified interactions.

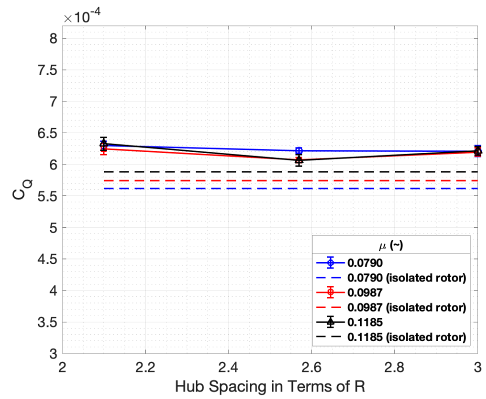
A comparison of the thrust coefficients between the plus and X configurations shows that there was a larger decrease in aft rotor performance for the plus configuration at $3R$; see Figure 18. While the shown aft rotor for both of these configurations was subjected to interactions from three other rotors, the aft rotor in the plus configuration was located downstream from all three other rotors, while the left aft rotor in the X configuration was operating directly downstream of only one of the upstream rotors. Thus, it was expected that the plus configuration would demonstrate lower thrust output for any given test condition, which it did. Furthermore, as expected, the plus configuration showed a larger decrease in thrust coefficient with decreasing hub spacing when compared to the X configuration.

Comparison to Flight Test Data

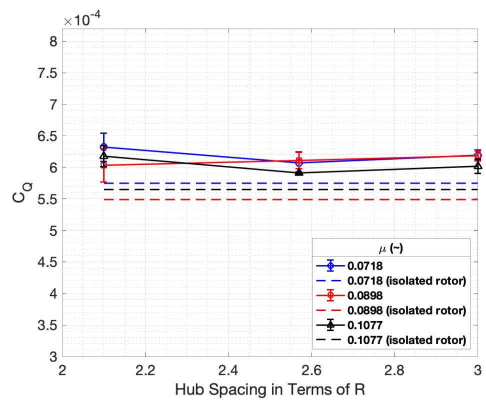
Figure 19 shows the plus and X configurations with the force/torque sensors numbered. Figure 20 shows the thrust coefficients for all four rotors in the plus configuration for both the wind tunnel experiments and the OSU flight tests. It can be seen that the front rotor produced the most thrust and the aft rotor produced the least, as expected. Additionally, it is evident that the two side rotors produced less thrust than the front rotor, more thrust than the back rotor, and equal thrust as each other. Figure 21 shows the thrust coefficients for all four rotors in the X configuration for both the wind tunnel experiments and the OSU flight tests. It can be seen that the front rotors produced more thrust than the aft rotors, as previously stated. Additionally, both front rotors produced equal thrust, as did both aft rotors. All of these results followed the expected trends.



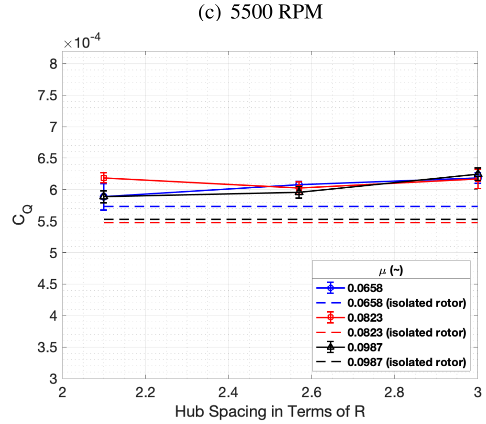
(a) 4500 RPM



(b) 5000 RPM



(c) 5500 RPM



(d) 6000 RPM

Figure 17. Torque coefficient for the left aft rotor in the X configuration for various hub spacings, advance ratios, and rotational frequencies

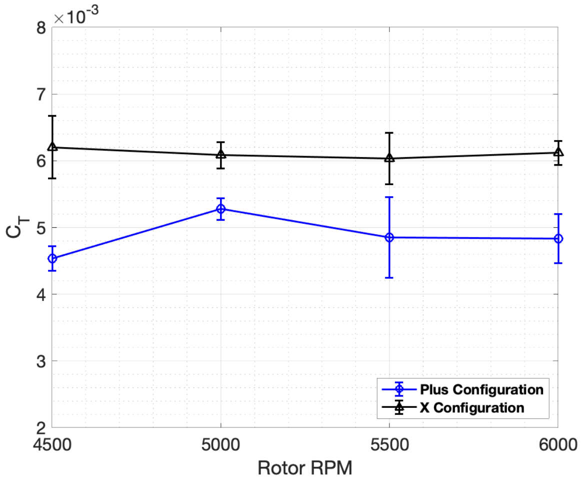


Figure 18. Thrust coefficients of the aft rotors in the plus and X configurations for a hub spacing of 3R, a wind tunnel speed of 8 m/s, and various rotational frequencies

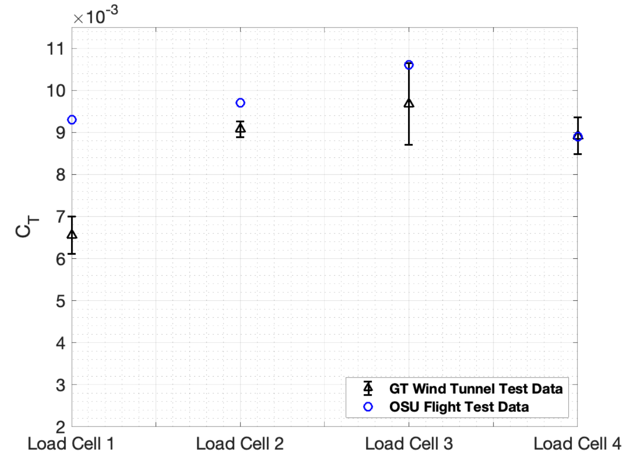


Figure 20. Thrust coefficients of all four rotors in the plus configuration for a hub spacing of 3.63R, a wind tunnel speed of 10 m/s, and a rotational frequency of 5000 RPM compared to the OSU flight test data (Ref. 24)

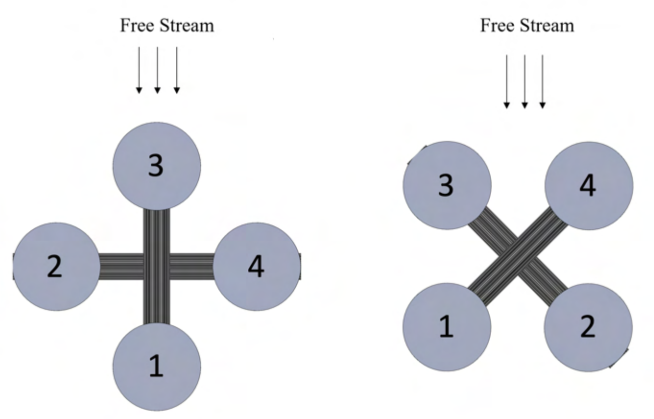


Figure 19. Schematic showing plus and X configurations with the force/torque sensors numbered

The data obtained from the OSU flight tests is published in (Ref. 24). This data includes the rotor thrust and torque of the multi-rotor vehicle in the plus and X configurations and for multiple skew angles relative to the free stream velocity. The motor and rotors used in the flight tests were identical to those used in the wind tunnel experiments. In addition, some of the wind tunnel test conditions were selected to mimic those experienced during the flight tests. Specifically, the rotational speeds of the rotors and the wind speed were selected based on the flight test conditions. Despite these similarities, there were some differences in the implementation of flight tests and wind tunnel experiments. First, the flight tests were carried out in trimmed conditions. In this case, a forward flight speed of about 10 m/s was maintained in every flight condition and the rotational speed of the individual rotors was allowed to vary to ensure this. In contrast, during the wind tunnel tests both the rotational speed of the rotors and the wind speeds were kept constant. In addition, the flight tests included the effects of the airframe of the flight test vehicle, while the wind tunnel tests included the potential influence of the rotor stands but no fuselage. Therefore, differences were expected in the observed aerodynamic interactions that occur in these flight and ground tests.

In spite of these differences, similar trends were observed. The thrust data obtained by OSU's flight tests are shown in Figures 20 and 21. It was observed from both flight test data and wind tunnel experiments that the aft-most rotor in the plus configuration and both the aft rotors in the X configurations produced lower thrust coefficients than the front rotors. This was caused by the aerodynamic interactions of the wake and flow structures generated by the front rotors with the aft rotor. Some differences in the flight test data and the wind tunnel tests were also observed. In the flight test data, the magnitudes of the thrust coefficients produced by the aft rotors were 39.8% and 30.0% larger than those produced by the equivalent rotor in the wind tunnel experiments in the X configuration, and 41.9% larger for the plus configuration; see Figures 20 and 21. This could be due to the differences in the atmospheric flow environment and trim conditions at which both tests were carried out. However, despite this difference in magnitudes, the trends were consistent between the wind tunnel and the flight tests.

Examining the front rotors in both X and plus configurations, wind tunnel measurements and flight test data were in good agreement, with differences of only 7.9% and 14.2% for the X configuration, and 9.5% for the plus configuration; see Figures 20 and 21. One peculiar difference was that the aft rotors in the X configuration produced less thrust than the aft-most rotor in the plus configuration in the flight tests, which was the opposite trend in the wind tunnel results. Apart from rotor trim effects (discussed previously) this difference might be attributed to the presence of the fuselage in the plus configuration, which can impede some interactions that may otherwise

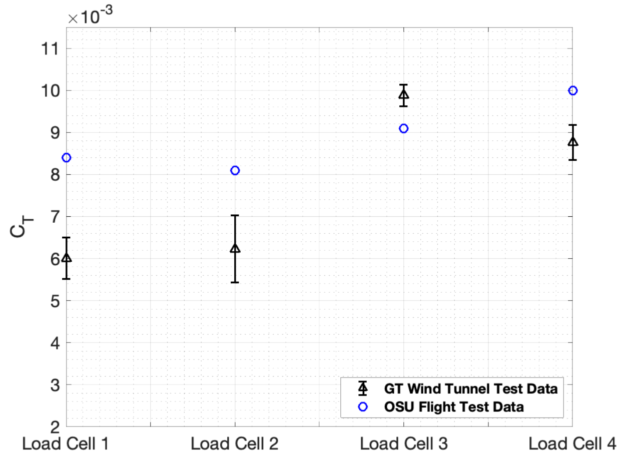


Figure 21. Thrust coefficients of all four rotors in the X configuration for a hub spacing of 2.6R, a wind tunnel speed of 10 m/s, and a rotational frequency of 5000 RPM compared to the OSU flight test data (Ref. 24)

occur from the front rotor. In the X configuration, however, the fuselage, that is not present in the test rig, has less of an effect on the interactions between the front and aft rotors; see Figure 19.

CONCLUSIONS

In this paper, an investigation of the aerodynamic interactions of rotors in multi-rotor vehicles was performed. Specifically, the effects that these interactions have on the performance (thrust and torque) of the individual rotors were investigated. Experiments were performed using force and torque measurements on all individual rotors in different multi-rotor configurations. Flow visualization and limited particle image velocimetry (PIV) was also performed to gain insights into the flow fields causing the modified rotor loads. The performance and flow visualization data was collected at a set of varying rotor rotational speeds from 4500RPM to 6000RPM, wind speeds of 8, 10, and 12 m/s, and hub spacings ranging from 2.1R to 3.63R, depending on the specific configuration. All experiments were preformed in the John J. Harper Wind Tunnel at the Georgia Institute of Technology. The following specific conclusions have been drawn.

1. For all of the investigated configurations (X, plus, and tandem) and hub spacings, it was found that the thrust of the aft rotors was reduced compared to the isolated rotor that was tested, which was an effect of the wake interactions and modified inflow on the aft rotors.
2. For all configurations, more torque was required for the aft rotors to maintain the rotational speed when compared to the isolated rotor.
3. As the hub spacing increased, wake interactions were less intense and the performance of the aft rotor improved. For the largest hub spacing, the aft rotors almost recovered the thrust of the isolated rotor for all of the configurations that were examined.
4. For a given advance ratio, it was found that the wake skew angle of the aft rotor increased as the hub spacing increased, because the wake interaction gradually decreased.
5. The increase in wind speed had less of an effect on the wake skew angles at lower hub spacings, because the wake effects from the front rotor on the inflow and downwash from the aft rotor offset the effects from the increase in streamwise velocity.
6. Between the plus and X configurations, the aft rotor thrust decreased more in the plus configuration, with an average C_T value of $1.4x10^{-3}$ lower than that of the aft rotors in the X configuration. This result was attributed to the additional interactions from the side rotors that the plus configuration aft rotor sees.
7. Good correlation was found between the OSU flight tests and the wind tunnel tests for the front and side rotors in X and plus configurations (7.9–14.2% difference), but a larger difference of 30–41.9% was found for the aft rotors, which is likely due to the different rotor trim conditions.

Future work will involve more flow field measurements using PIV to get further insights into the specific wake and vortex interactions that are causing the modified steady loads. Furthermore, the vibratory loads that were measured but not shown in this work will be analyzed and correlated to the unsteady vortical wake interactions between rotors in multi-rotor vehicles.

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CONTACT INFORMATION

All the authors work at the Georgia Institute of Technology in the Daniel Guggenheim School of Aerospace Engineering. Contact emails are listed below.

aatte3@gatech.edu

daley.wylie@gatech.edu

juergen.rauleder@gatech.edu

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